Homework 4

October 16, 2026

Problem 1

Find out the Smith canonical form of the λ-matrix

$$A(\lambda) = \begin{pmatrix} 1-\lambda & \lambda^2 & \lambda \\ 2\lambda - 1 & \lambda(1-\lambda) & -2\lambda \\ 1+\lambda^2 & \lambda^2 & -\lambda^2 \end{pmatrix}.$$

Problem 2

Let $A(\lambda)$ be the $n \times n$ λ-matrix

$$A(\lambda) = \begin{pmatrix} \lambda - a & b_1 & 0 & \cdots & 0 \\ 0 & \lambda - a & b_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda - a & b_{n-1} \\ 0 & \cdots & \cdots & 0 & \lambda - a \end{pmatrix},$$

where $b_1, b_2, \ldots, b_{n-1}$ are all nonzero constants. Find out the invariant polynomials of $A(\lambda)$.

Problem 3

Find out the Smith canonical form of the λ-matrix

$$A(\lambda) = \begin{pmatrix} \lambda & 0 & \cdots & 0 & a_0 \\ -1 & \lambda & \ddots & & a_1 \\ 0 & -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \lambda & a_{n-2} \\ 0 & \cdots & 0 & -1 & \lambda + a_{n-1} \end{pmatrix}.$$

Problem 4

With respect to a matrix $A \in F^{n \times n}$ and an indeterminate λ, $\lambda E - A$ is called the eigenmatrix, where E is the identity matrix. Find out the Smith canonical form of the eigenmatrix of the following matrix

$$A = \begin{pmatrix} -7 & -12 & -6 \\ 3 & 5 & 3 \\ 3 & 6 & 2 \end{pmatrix}.$$